DNV·GL

DVR-PP127937
Rev.0

DESIGN VERIFICATION REPORT

Designer/ Manufacturer	: Special and Safety Work (SSW), Riyadh, Saudi Arabia.
DNV GL Project No	: PP127937
Equipment Description	: High pressure blast resistant door (Sliding)
Codes and Standards	: AISC (9th Edition)

This is to confirm that the structural design of the High pressure blast resistant sliding door (as per drawings listed in Table-1:Design Examination Document) has been reviewed and found to be in compliance with the applicable codes and standards mentioned in this Design Verification Report.

Scope of Verification:

The verification of the design of Door is limited to review of structural analysis submitted in compliance with AISC (9th Edition) as applicable with regrads to local in-place strength against uniformly distributed static equivalent peak over pressure of 10 barg.

No other codes / standards have been referred to and it is concluded that client would satisfy themselves that requirements of other relevant codes / standards / national regulations e.g. those mentioned as reference in design calculations submitted etc., are complied with suitably.

DNV GL review is limited to the door itself. This review does not cover the design of concrete supports around the door.

Table-1: Design Examination Document:

S. No.	Drawing No.	Revision	Drawing Title	Status
1.	SSW-FSC-01 Sheet 1 of 3	03	High Pressure Blasr Resistant Door – Plan elevation and Details	Reviewed
2.	SSW-FSC-01 Sheet 2 of 3	03	High Pressure Blasr Resistant Door – Plan elevation and Details	Reviewed

Verification Limitations:

Design Parameters:

Parameter	Description
Blast Pressure	10 barg
Overall Door Size	1600mm width and 2320mm height

Other design data are considered as per drawing listed in Table-1.

Limitations and Notes:

(1) The design appraisal is limited to the static in place analysis of the steel door frame and any aspects towards adequacy of anchoring arrangement, inspection and testing shall be ensured by the manufacturer.

DET NORSKE VERITAS 1864 ABU DHABI

(2) Door structural steel must have yield strength not less than 250 MPa as used in design calculation. All material used for door and the hardware components like door frame anchors, tower bolts, hinges, etc. including sealants along with fabrication and installation arrangements shall meet the blast rating specifications and structural quality as per the client requirements.
(3) Present appraisal has been carried out with the blast load indicated in the submitted calculations. Other loads/ load factors like brittle fracture, etc are not considered in this reivew.

Validity Period:

No validity period is stated on the design verification report as it only confirms the conformity with specified regulations / standards at the time of issue. Change in regulations or change in design may have influence on the validity.

Signed for **Det Norske Veritas AS, Abu Dhabi Branch**

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for Rajeev Sharma
Discipline Lead - Static, Container and Lifting Appliance

Balaji Selvaraj
Mechanical Engineer - VMC

OPMAE 646

Date: 08th March 2015
Place: Dubai, UAE